

Measuring Electromagnetic Induction: The Peak EMF Induced by a Magnet Falling Through a Coil

Chhaya Ramnani (u8336188)

Lab partners: Tsz Wing Candice Ng, Chinatsu Komada
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1.0 Abstract

This experiment examined how the peak electromotive force (emf) induced by a magnet falling through a coil depends on three things: the number of coil turns N , the coil radius R , and the magnet's speed s as it passed through. The magnet's dipole moment was first found from calibration measurements of its on-axis field, giving $m = (1.18 \pm 0.07) \text{ A m}^2$. This was used to build a theoretical prediction for the peak emf with no free parameters, which the measured data from each of the three sub-experiments was then compared against directly — no scaling factor was fitted anywhere. The peak emf rose roughly linearly with N and with s , and fell off roughly as R^{-2} with coil radius, matching the predicted shape $\varepsilon_{\text{peak}} \propto Ns/R^2$. None of the three datasets matched the absolute theoretical prediction within uncertainty, though, and the N data disagreed by far the most (a measured-to-theory ratio of 0.461 ± 0.003). This was mainly put down to the magnet's finite size breaking the point-dipole assumption the theory relies on, and, specifically for the N sub-experiment, to an unresolved mismatch between two setups that should have given the same reading.

2.0 Introduction

2.1 Aims

This experiment set out to see how the peak emf induced by a magnet falling through a coil depends on three parameters: the number of turns in the coil, N ; the coil radius, R ; and the magnet's speed as it passed through, s . A dipole model of the magnet was calibrated first from independent field measurements, and the resulting zero-parameter theoretical prediction was then compared directly against the measured emf for each sub-experiment, with no scaling factor fitted in.

The peak emf was expected to increase linearly with N , increase linearly with s , and fall off as R^{-2} with coil radius, with all three trends together matching the combined proportionality $\varepsilon_{\text{peak}} \propto Ns/R^2$.

2.2 Background theory

By Faraday's law, the emf induced in a circuit equals the negative rate of change of magnetic flux Φ_B through it:

$$\varepsilon = -\frac{d\Phi_B}{dt}, \quad \Phi_B = \int B_{\perp} dA. \quad (1)$$